

CaptionAI: AI-Powered Social Media Caption Generator

Project Description:

The Human Development Index (HDI) is a statistical composite index of life expectancy, education, and per capita income indicators, which are used to rank countries into four tiers Very High, High, Medium, and Low of human development. A country achieves a higher HDI when its population enjoys a longer lifespan, better educational attainment, and a higher Gross National Income (GNI PPP) per capita.

The HDI was developed to emphasize that people and their capabilities should be the primary measure of a country's development rather than economic growth alone. It provides a broader perspective on development by considering health, education, and living standards together. The index is widely used by governments, researchers, policymakers, and international organizations to evaluate progress, compare countries, and identify areas requiring improvement.

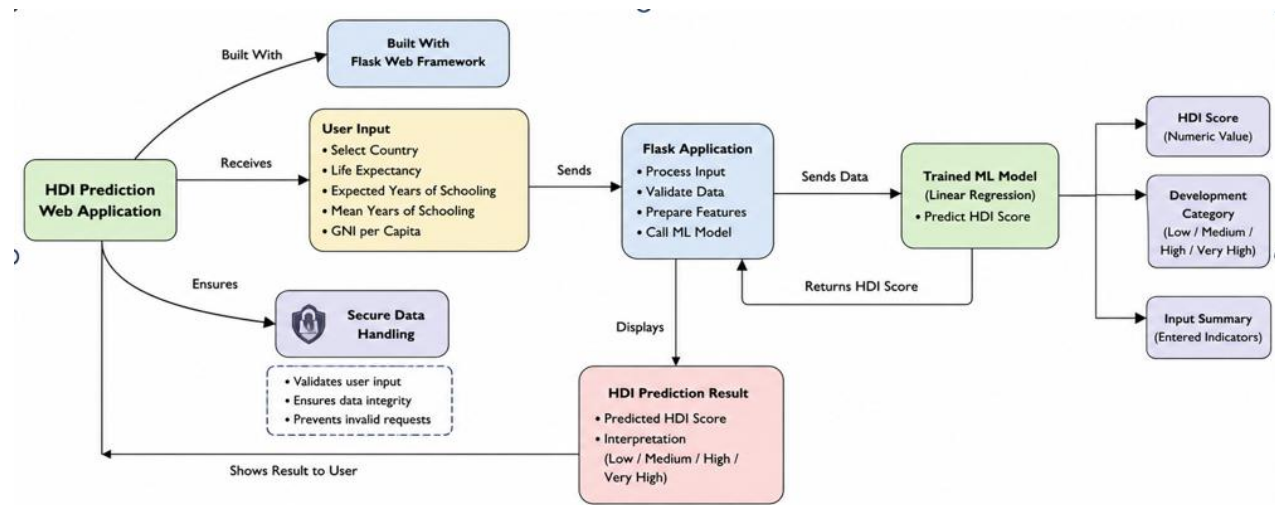
Scenarios

Scenario 1: A user selects a country with high life expectancy, strong mean years of schooling, expected years of schooling, and a high GNI per capita. Based on these indicators, the model predicts a Very High HDI score, classifying the country among the most developed nations. This helps users understand how strong performance across multiple development dimensions contributes to overall human development.

Scenario 2: A policymaker inputs mid-range values representing a developing nation, including moderate life expectancy, average educational attainment, and a moderate income level. The model returns a Medium HDI score, providing insights into areas where improvements in healthcare, education, or income generation could significantly enhance human development outcomes.

Scenario 3: A researcher evaluates a country with relatively low life expectancy, limited educational opportunities, and low GNI per capita. The model predicts a Low HDI score and highlights the country's developmental challenges. Such predictions can support governments and development organizations in prioritizing investments and policy interventions.

Technical Architecture:



Description: The Human Development Index (HDI) Prediction System follows a three-tier architecture consisting of the presentation, application, and machine learning layers. The frontend is developed using HTML and CSS to provide a user-friendly interface. The backend is built using the Flask framework to process user requests. A trained Linear Regression model is used to predict the HDI score based on user inputs. The model is stored as a pickle (.pkl) file and loaded by the Flask application. The predicted HDI score is displayed to the user through the web interface. This architecture provides a simple, efficient, and reliable prediction system.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

Pre-requisites:

- **Flask Framework Knowledge:** [Flask Documentation](#)
- **Groq API Familiarity:** [Groq API Documentation](#)
- **HTML, CSS, and JavaScript Skills:** [W3Schools HTML/CSS/JavaScript Tutorials](#)
- **Python Programming Proficiency:** [Python Documentation](#)
- **Version Control with Git:** [Git Documentation](#)
- **Development Environment Setup:** [Flask Installation Guide](#)
- **Ngrok for Public Deployment:** [Ngrok Documentation](#)

Project Workflow:

Milestone 1: Dataset Collection and Model Development

- Activity 1.1: Collect the Human Development Index (HDI) dataset from a reliable source and load it into the development environment.
- Activity 1.2: Analyze and preprocess the dataset by selecting the required features, handling missing values, and preparing the data for training.
- Activity 1.3: Select the Linear Regression algorithm for predicting the Human Development Index (HDI).
- Activity 1.4: Set up the development environment by installing the required Python libraries, Flask framework, and machine learning dependencies.

Milestone 2: Machine Learning Model Development

- Activity 2.1: Split the dataset into training and testing datasets for model development.
- Activity 2.2: Train the Linear Regression model using the training dataset and evaluate its performance.
- Activity 2.3: Save the trained model as HDI.pkl using the Pickle library for future predictions.

Milestone 3: App.py Development

- Activity 3.1: Develop the main application logic in app.py, including loading the trained model, processing user inputs, defining Flask routes, and generating HDI predictions.

Milestone 4: Frontend Development

- Activity 4.1: Design and develop the user interface using HTML and CSS, ensuring a simple, responsive, and user-friendly layout.
- Activity 4.2: Create Flask templates using render_template() to display the Home page, Prediction page, and Result page based on user interactions.

Milestone 5: Deployment

- Activity 5.1: Prepare the Flask application for local deployment by organizing the project structure and installing all required dependencies.
- Activity 5.2: Run, test, and validate the application on the local Flask server to ensure accurate HDI predictions and smooth functionality.

Milestone 6: Conclusion

- Activity 6.1: Summarize the overall project, highlighting the successful development of the Human Development Index (HDI) Prediction System using Machine Learning and Flask.
- Activity 6.2: Discuss the project's future improvements, acknowledge the successful completion of all objectives, and conclude the project demonstration.

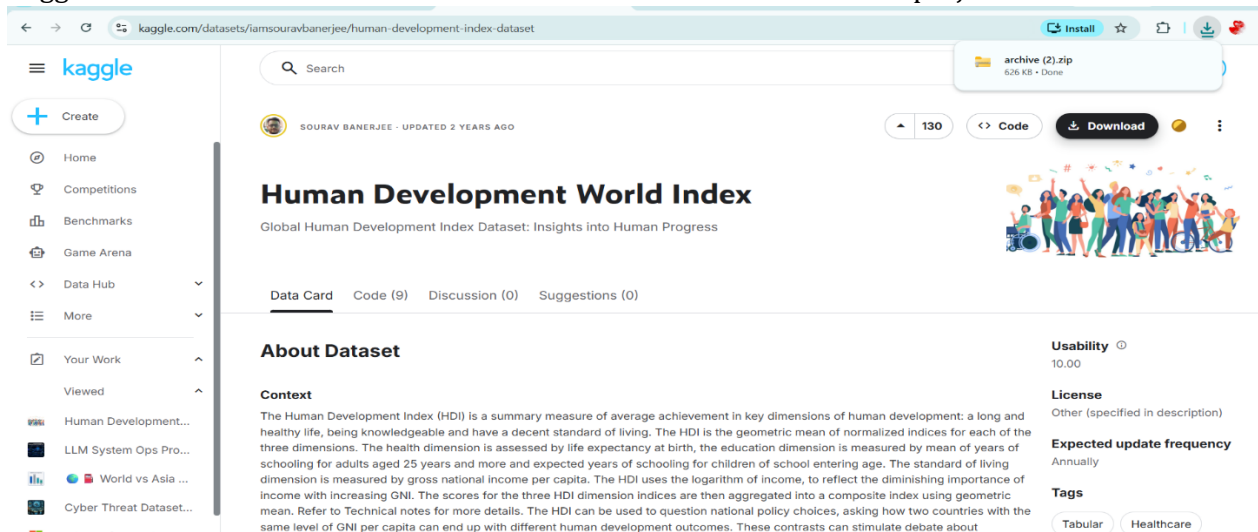
Milestone 1: Dataset Collection and Model Development

In this milestone, the Human Development Index (HDI) dataset is collected and analyzed. The required features are identified, and the Linear Regression algorithm is selected for predicting the HDI score. The development environment is also prepared by installing the necessary Python libraries and Flask framework.

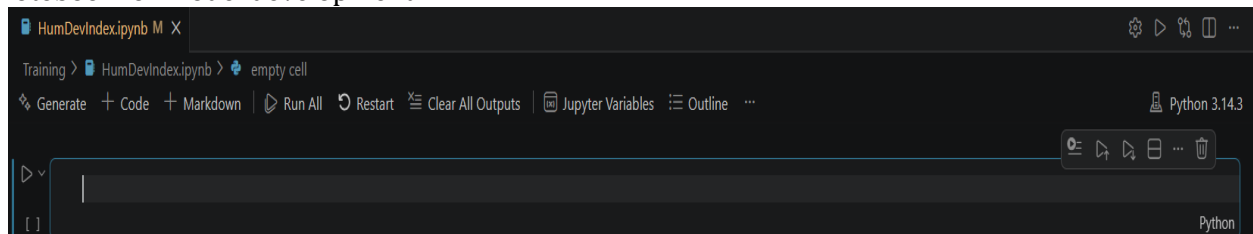
Activity 1.1: Download and Load the Dataset

- Before training the model, download the HDI dataset and load it into Jupyter Notebook.

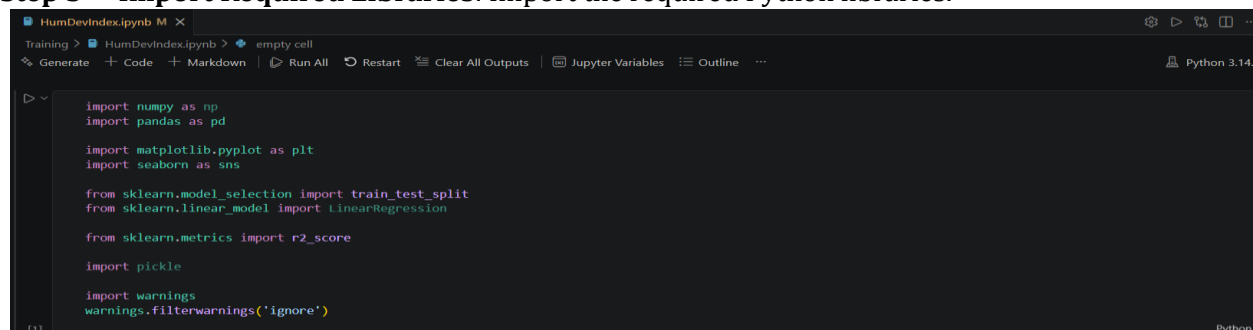
Step 1 – Download the Dataset: Download the Human Development Index (HDI) dataset from Kaggle or another reliable source and save it in the **Dataset** folder of the project.



Step 2 — Step 2 – Open Jupyter Notebook: Launch Jupyter Notebook and create a new Python Notebook for model development.



Step 3 —Import Required Libraries: Import the required Python libraries.



Step 4 — Load the Dataset: Read the dataset using Pandas.

```
df = pd.read_csv("../Dataset/HDI.csv")
```

Step 5 — Display the Dataset: View the first few records to understand the dataset.

Step 6 — Check Dataset Information: Display column names, data types, and missing values.

Activity 1.2: Select the Machine Learning Algorithm

The next step is to select a suitable Machine Learning algorithm for predicting the Human Development Index.

- Understand the Project Requirements: Identify the input features and target variable required for prediction.
- Analyze the Dataset: Study the relationship between Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, Gross National Income, and HDI.
- Select the Algorithm: Choose the Linear Regression algorithm because the project predicts a continuous numerical value (HDI score).

Activity 1.3: Define the Application Architecture

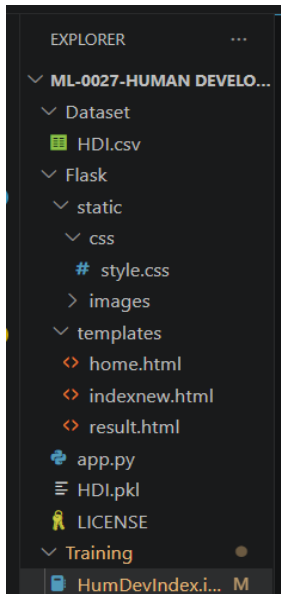
- Design the System Architecture: Create a diagram showing the interaction between the HTML/CSS frontend, Flask backend, Machine Learning model, and dataset.
- Frontend: Design the Home, Prediction, and Result pages using HTML and CSS.
- Backend: Develop the Flask application to receive user inputs and communicate with the trained model.
- Machine Learning Model: Use the trained Linear Regression model to predict the HDI score and return the result to the user.

Activity 1.4: Set Up the Development Environment

- starting development, install all the required software and libraries.
- **Step 1 – Install Python:** Install Python 3.10 or above.
- **Step 2 – Install Required Libraries:**

```
Microsoft Windows [Version 10.0.26200.8737]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\K.Prasanna\OneDrive\Desktop\ML-0027-Human Development Index>pip install numpy pandas matplotlib seaborn scikit-learn flask
```

- **Step 3 – Install Visual Studio Code:** Use Visual Studio Code to develop the Flask application.
- **Step 4 – Organize the Project Structure:** Create folders such as Dataset, Training, Flask, Templates, Static, and Model.
- **Step 5 – Verify the Installation:** Run the Python program and ensure that all required libraries are imported successfully without errors.
- **Set Up Application Structure:** Create the project directory structure by organizing the project into separate folders for dataset, training notebook, HTML templates, static files (CSS and images), and the Flask application files.



Milestone 2 Machine Learning Model Development

Activity 2.1: Develop the Machine Learning Model

Implement the core functionality of predicting the Human Development Index (HDI) using the Linear Regression algorithm.

- Data Preprocessing: Clean the dataset and select the required input and output features.
- Model Training: Train the Linear Regression model using the prepared training dataset.
- Model Evaluation: Test the model using the testing dataset and evaluate its prediction accuracy.
- Save the Model: Store the trained model as HDI.pkl using the Pickle library for future predictions.

Activity 2.2: Implement the Flask Backend

Develop the Flask backend to process user inputs and generate HDI predictions.

Define Routes in Flask:

- Create routes for the **Home Page**, **Prediction Page**, and **Result Page** in app.py.

Process User Input:

- Create a form in **indexnew.html** to collect the country name, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI).
- Use Flask's request.form to retrieve the values entered by the user.
- Validate the input values before sending them to the prediction model.

Generate Prediction:

- Load the trained **HDI.pkl** model.
- Pass the user input to the model for prediction.
- Display the predicted HDI score on the **Result Page**.

Milestone 3: App.py Development

Milestone 3 focuses on developing the **app.py** file, which is the main component of the Human Development Index Prediction System. In this milestone, Flask routes are created to connect the web pages with the Machine Learning model. The application receives user inputs, processes them using the trained model, and displays the predicted HDI score.

Activity 3.1: Writing the Main Application Logic in app.py

Define the Core Routes in app.py

Create the following routes in the Flask application:

- `/` → Displays the Home page (`home.html`).
- `/predict` → Opens the Prediction page (`indexnew.html`).
- `/predict_result` → Receives user input, loads the trained model, predicts the HDI score, and displays the result (`result.html`).

```
    ) !DOCTYPE html>
<html>
<head>
    <title>Human Development Index</title>

    <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
</head>

<body>

<div class="container">

    <h1>Human Development Index</h1>

    <h3 style="color:white;">
        HDI is used to measure the overall development of a country based on health, education, and
        income. It helps compare countries, identify development gaps, and support better policy and planning
        decisions.
    </h3>

    <p style="color:#E6E6E6; font-size:16px; line-height:28px; margin-top:20px;">
        This application predicts the Human Development Index (HDI)
        using four important indicators:
        <br><br>
        • Life Expectancy at Birth
        <br>
        • Expected Years of Schooling
        <br>
        • Mean Years of Schooling
        <br>
        • Gross National Income Per Capita
        <br>
    </p>

    <br>

    <a href="/predict">
        <button class="btn">Start Prediction</button>
    </a>

</div>

</body>
</html>
```


Process User Input

- Create an input form in **indexnew.html** to collect:
 - Country Name
 - Life Expectancy
 - Expected Years of Schooling
 - Mean Years of Schooling
 - Gross National Income (GNI)
- Use **request.form** to receive the values entered by the user.
- Validate the input values before sending them to the prediction model.

```
} <!DOCTYPE html>
<html>
<head>
  <title>Human Development Index</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
</head>

<body>

<h1 align="center">Human Development Index</h1>

<div class="container">

<form action="/predict_result" method="POST">

<select name="country" required>
  <option value="">Select the Country</option>

  {% for country in countries %}
    <option value="{{ country }}">{{ country }}</option>
  {% endfor %}
</select>

<br>
```

Load the Trained Model

- Load the saved **HDI.pkl** model using the Pickle library.
- Convert the user input into the required format.
- Pass the input values to the trained Linear Regression model.

Generate the Prediction

- Predict the Human Development Index (HDI) score using the trained model.
- Classify the predicted score into one of the following categories:
 - Very High HDI
 - High HDI
 - Medium HDI

- Low HDI

Display the Result

- Send the predicted HDI score and category to **result.html**.
- Display the prediction result on the Result page with a **Predict Again** button.

```
} <!DOCTYPE html>
<html>
<head>
  <title>Prediction Result</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
</head>

<body>

<div class="container">

<h1>Human Development Index</h1>

<h3 style="color:gold;">
A machine learning web application using Flask
</h3>

<br>

<h2 style="color:yellow;">

{% if prediction >= 0.800 %}
Very High HDI : {{ prediction }}

{% elif prediction >= 0.700 %}
High HDI : {{ prediction }}

{% elif prediction >= 0.550 %}
Medium HDI : {{ prediction }}

{% else %}
Low HDI : {{ prediction }}

{% endif %}

</h2>

<br>

<a href="/predict">
  <button>Predict Again</button>
</a>

</div>

</body>
</html>
```

Milestone 4: Frontend Development

Milestone 4 focuses on designing a simple, user-friendly, and responsive interface for the Human Development Index Prediction System. HTML and CSS are used to create attractive web pages, while Flask templates are used to display user inputs and prediction results.

Activity 4.1: Designing and Developing the User Interface

Set Up the Base HTML Structure:

- Develop three HTML pages:
 - **home.html** – Displays the project introduction and Start Prediction button.
 - **indexnew.html** – Allows users to enter the required HDI details.
 - **result.html** – Displays the predicted HDI score and category.
- Use semantic HTML elements to organize the web pages.
- Add headings, labels, buttons, and input fields for better user interaction.

Design a Responsive Layout Using CSS

- Create a **style.css** file to design all the web pages.
- Apply a background image, colors, fonts, and button styles to improve the appearance.
- Design responsive pages that work properly on different screen sizes.
- Style the input fields, prediction button, and result page for a clean and attractive interface.

Create User Input Forms

- Create input fields for:
 - Country Name
 - Life Expectancy
 - Expected Years of Schooling
 - Mean Years of Schooling
 - Gross National Income (GNI)
- Add a **Predict** button to submit the values.
- Display the predicted HDI score and category on the Result page.

Activity 4.2 : Integrating Frontend with Flask

Handle User Input

- Connect the HTML form with the Flask backend using the POST method.
- Send the entered values to the Flask application for prediction.

Display Prediction Result

- Receive the predicted HDI score from the backend.
- Show the prediction result on result.html.
- Add a Predict Again button to return to the prediction page.

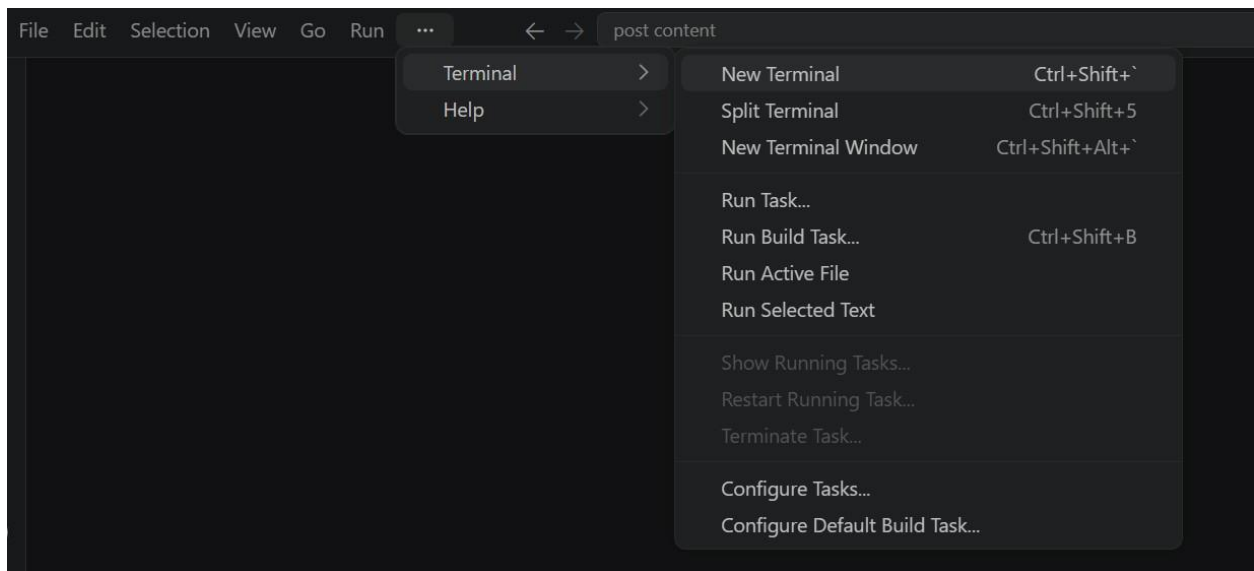
Milestone 5: Deployment

In Milestone 5, the Human Development Index Prediction System is prepared for deployment. The Flask application is tested locally to ensure that the prediction model, user interface, and backend work correctly. The application is then made ready for demonstration and public access if required

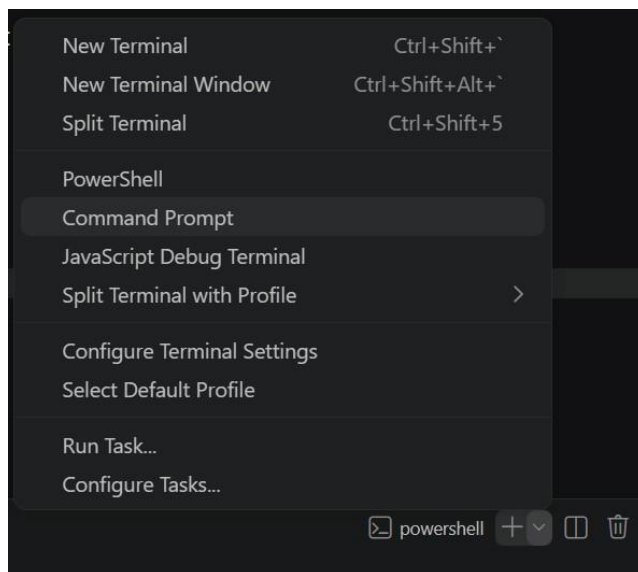
Activity 5.1: Preparing the Application for Local Deployment

Set Up a Virtual Environment:

- Open a New Terminal



- Then open "Command Prompt"



Set Up the Development Environment

- Install Python and the required libraries.
- Open the project folder in Visual Studio Code.
- Ensure that all project files, including **app.py**, **HDI.pkl**, HTML templates, CSS files, and images are placed in their respective folders.

Run the Flask Application

- Open the terminal.
- Navigate to the Flask project folder.
- Run the application using the following command:

Activity 5.2: Local Testing and Verification

Start the Flask Development Server:

- Run the application locally using:
- Once running, open a browser and navigate to “http://127.0.0.1:5000” to access the app.

```
C:\Users\K.Prasanna\OneDrive\Desktop\ML-0027-Human Development Index\Flask>python app.py
Current Working Directory: C:\Users\K.Prasanna\OneDrive\Desktop\ML-0027-Human Development Index\Flask
App Root Path: C:\Users\K.Prasanna\OneDrive\Desktop\ML-0027-Human Development Index\Flask
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
Current Working Directory: C:\Users\K.Prasanna\OneDrive\Desktop\ML-0027-Human Development Index\Flask
App Root Path: C:\Users\K.Prasanna\OneDrive\Desktop\ML-0027-Human Development Index\Flask
* Debugger is active!
* Debugger PIN: 605-784-029
```

Test the Application

- Click the **Start Prediction** button on the Home page.
- Enter the required values:
 - Country Name
 - Gross National Income (GNI)
 - Expected Years of Schooling
 - Mean Years of Schooling
 - Life Expectancy
- Click the **Predict** button.

- Verify that the predicted HDI score and HDI category are displayed correctly on the Result page.
- Test the application using different country values to ensure accurate predictions.

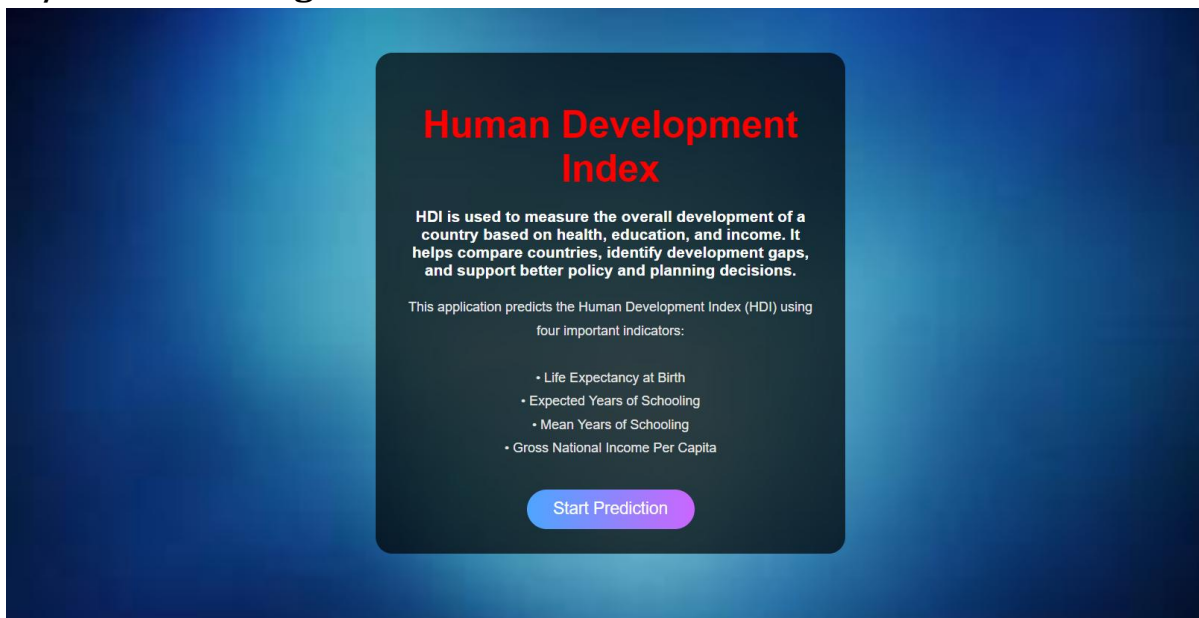
Activity 5.3: Project Demonstration

Demonstrate the Web Application

- Open the application in a web browser.
- Explain the purpose of the Human Development Index Prediction System.
- Demonstrate the complete workflow:
 - Home Page
 - Prediction Page
 - Result Page
- Show how different input values produce different HDI prediction results.
- Verify that all pages are functioning correctly without errors.

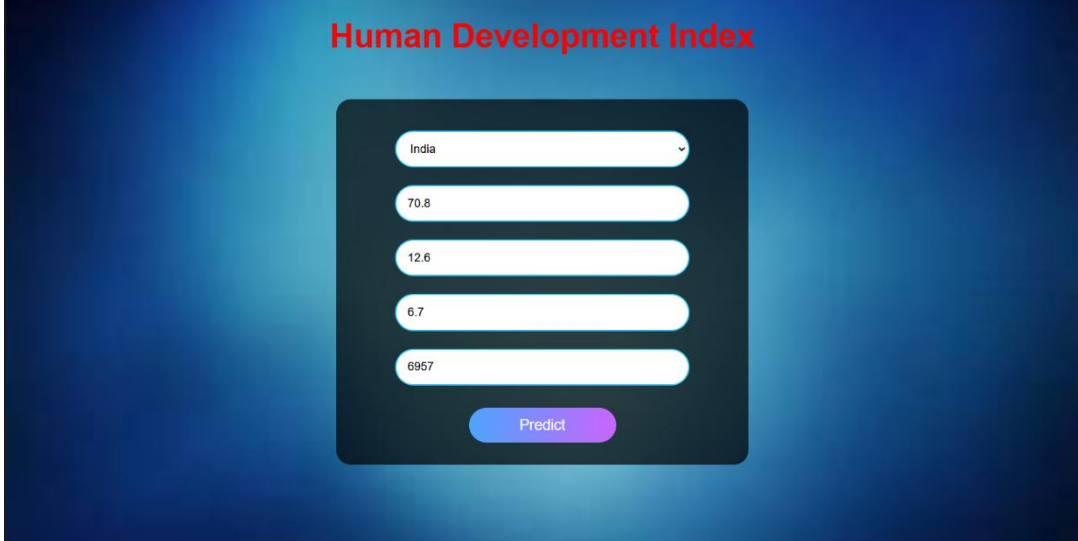
Exploring the Website's Web Pages:

Home / Generator Page:



Description: The Home page introduces the Human Development Index Prediction System. It provides a brief explanation of HDI and its importance. Users can start the prediction process by clicking the Start Prediction button.

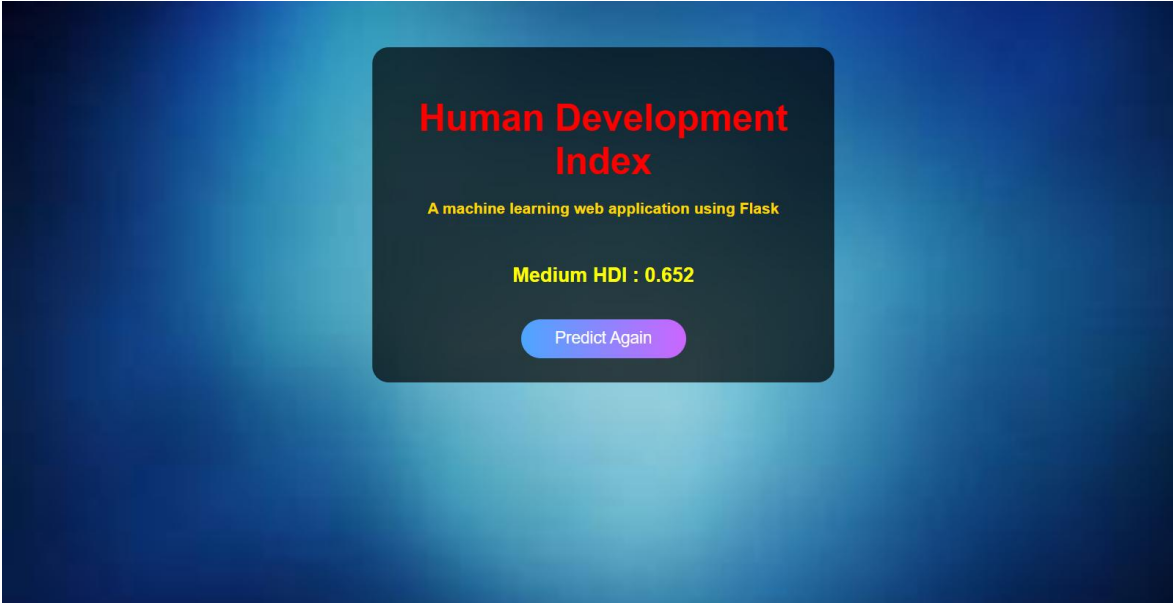
Input Section:



The input form is titled "Human Development Index" in red. It contains five input fields: a dropdown menu for "India", and four text boxes containing "70.8", "12.6", "6.7", and "6957". A purple "Predict" button is at the bottom.

Description: The Prediction page contains a user-friendly form where users enter the required information, including Country Name, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI). After entering the values, users click the Predict button to generate the HDI prediction.

Results Section:



The results page is titled "Human Development Index" in red. Below the title, it says "A machine learning web application using Flask" in yellow. The predicted score is displayed as "Medium HDI : 0.652" in yellow. A purple "Predict Again" button is at the bottom.

Description: The Result page displays the predicted Human Development Index (HDI) score along with its corresponding category (Very High, High, Medium, or Low HDI). A Predict Again button is provided so users can return to the Prediction page and test different input values.

Conclusion:

The Human Development Index (HDI) Prediction System is a machine learning-based web application that predicts a country's HDI using key indicators such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per capita. Built using Flask and Linear Regression, the application provides a simple and user-friendly interface for making predictions. This project demonstrates how machine learning can support data-driven analysis and help users better understand human development across countries. In the future, the system can be enhanced by using real-time datasets, advanced prediction models, and interactive visualizations to improve its accuracy and usability.